 Name _____ Date _____

Your Body's Home Team: The Microbiome

How do helpful bacteria keep you healthy — and what happens when they disappear?

You carry about 38 trillion bacteria — roughly the same as the number of human cells in your body. Most of them are your allies. Today you'll explore your microbiome.

PART A The Gut as an Ecosystem

Read: *Your gut is an ecosystem. Just like a rainforest, a gut with more* ***biodiversity*** *(more kinds of bacteria) is more stable and harder for harmful bacteria to invade. Studies show that the Hadza people of Tanzania, who eat wild, varied foods, have about twice as many kinds of gut bacteria as most people in the U.S.*

1. What is biodiversity? Give one example from an ecosystem you know (not the gut).

2. How is a diverse gut microbiome similar to a diverse rainforest? What do both have in common?

3. Predict: what might happen to your gut microbiome if you ate the exact same three foods every day for a year?

PART B Compare Microbiomes

Use the information below to fill in the comparison table.

Healthy gut: *500+ kinds of bacteria · breaks down fiber · makes vitamins B and K · trains immune system · defense against pathogens* ***After strong antibiotics:*** *fewer than 100 kinds of bacteria · C. diff bacteria may move in · less vitamin production · immune system less trained · more vulnerable to infection*

Feature	Healthy gut microbiome	After strong antibiotics
Number of bacteria types		
Vitamins produced		
Risk of infection		
Immune system		
Biodiversity level		

PART C Your Gut Is Like a Volleyball Team

Your microbiome works together like a volleyball team. Match each gut bacteria job to its volleyball position.

Gut bacteria job	Volleyball position	Explain the connection
Take up all the space and food so harmful bacteria can't move in	Libero (back row defense)	
Break down fiber and make vitamins	Middle blocker	
Train the immune system to recognize enemies	Head coach	
Attack and crowd out invading pathogens	Outside hitter	

PART D Evaluate a Claim

> ***Claim from a probiotic yogurt ad:*** *"Our yogurt contains 10 billion live bacteria cultures. Taking our yogurt every day will improve your immune system and prevent you from getting sick."*

4. What part of this claim is supported by science? Use what you know about the microbiome.

5. What part of the claim might be an exaggeration or misleading? Explain.

6. What question would a scientist ask before accepting this claim?

PART E Mutualism: Both Sides Win

Mutualism is a relationship where two different organisms both benefit. Your relationship with your gut bacteria is one of the best examples in biology.

7. List TWO things you give your gut bacteria:

8. List THREE things your gut bacteria give you:

9. Extend it: Can you think of another mutualism relationship in nature? Describe what each organism gives and gets.

■ **Exit Ticket** A doctor prescribes you a strong antibiotic for a sinus infection. Using what you know about the microbiome, describe ONE potential side effect you might experience and explain why it happens.

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. Reference: learn.genetics.utah.edu/content/microbiome/ · Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html